DSP · L06

Spectral analysis and block convolution

Cheat sheet · Day 11 · Tue 3 Nov · pack D6

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

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SPECTRAL ANALYSIS (L06 s29-32)
 Fs >= 2B        Ts = 1/Fs        T0 = L*Ts = L/Fs   (L = real samples)
 Resolution  = 1/T0 = Fs/L        Bin spacing = Fs/N   (same if no zero-pad)
 Bin freq    F_k = k*Fs/N,  w_k = 2*pi*k/N;  k > N/2  ->  -(N-k)*Fs/N
 Real tone F0 -> bins k0 = F0*N/Fs and N-k0; non-integer k0 => leakage
 Design: Fs = 2B; N >= Fs/dF, round UP to 2^m; T0 = N/Fs; quote Fs/N
 Zero-padding: finer grid (interpolation) only, NO new resolution
 Aligned A*cos, rect, N = L: |X(k0)| = A*N/2, all other bins 0
 Hanning: w(n) = 0.5(1 - cos(2*pi*n/(L-1))), 0 <= n < L
   rect : main lobe 2 bins (4pi/L), sidelobe -13 dB
   hann : main lobe ~4 bins (8pi/L), sidelobe -31 dB, gain ~0.5
   => less leakage, worse resolution (budget ~2x the record length)

BLOCK CONVOLUTION (L06 s25-27)   N = L + M - 1, h zero-padded to N once
 OLA: blocks of L + (M-1) zeros; block*h = linear (exactly N long);
      place block i at n = iL; ADD the M-1 overlapping samples
 OLS: block = last M-1 of previous + L new (first block: M-1 zeros);
      circular = linear conv, then fold the last M-1 onto the first M-1;
      DISCARD the first M-1 outputs, keep L
 Blocks: OLA ceil(Lx/L);  OLS ceil((Lx+M-1)/L);  total output Lx+M-1
 T5 Q4 is OLS with M=2: n=0 wrong; fix with +0.5*y(63)
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